

and

$$\begin{aligned}
 W_p(\mu, \nu) &\leq \left(\int |x-z|^p d\gamma \right)^{\frac{1}{p}} = \left(\int |x-z|^p d\sigma \right)^{\frac{1}{p}} = \|x-z\|_{L^p(\sigma)} \\
 &\leq \|x-y\|_{L^p(\sigma)} + \|y-z\|_{L^p(\sigma)} = \left(\int |x-y|^p d\sigma \right)^{\frac{1}{p}} + \left(\int |y-z|^p d\sigma \right)^{\frac{1}{p}} \\
 &= \left(\int |x-y|^p d\gamma^+ \right)^{\frac{1}{p}} + \left(\int |y-z|^p d\gamma^- \right)^{\frac{1}{p}} = W_p(\mu, \rho) + W_p(\rho, \nu).
 \end{aligned}$$

The proof is concluded. $\square$ $\square$

The following lemma is usually known as “gluing lemma” and allows to produce a sort of “composition” of two transport plans, as if they were maps.

Lemma 5.5. *Given two measures $\gamma^+ \in \Pi(\mu, \rho)$ and $\gamma^- \in \Pi(\rho, \nu)$ there exists (at least) a measure $\sigma \in \mathcal{P}(\Omega \times \Omega \times \Omega)$ such that $(\pi_{x,y})_{\#}\sigma = \gamma^+$ and $(\pi_{y,z})_{\#}\sigma = \gamma^-$, where $\pi_{x,y}$ and $\pi_{y,z}$ denote the projections on the two first and two last variables, respectively.*

Proof. Start by taking γ^+ and disintegrate it w.r.t. the projection π_y (see Box 2.2 in Section 2.3). We get a family of measures $\gamma_y^+ \in \mathcal{P}(\Omega)$ (as we pointed out in the same box, we can think of them as measures over Ω , instead of viewing them as measures over $\Omega \times \{y\} \subset \Omega \times \Omega$). They satisfy (and are actually defined by)

$$\int_{\Omega \times \Omega} \phi(x, y) d\gamma^+(x, y) = \int_{\Omega} d\rho(y) \int_{\Omega} \phi(x, y) d\gamma_y^+(x),$$

for every measurable function ϕ of two variables. In the same way, one has a family of measures $\gamma_y^- \in \mathcal{P}(\Omega)$ such that for every ϕ we have

$$\int_{\Omega \times \Omega} \phi(y, z) d\gamma^-(y, z) = \int_{\Omega} d\rho(y) \int_{\Omega} \phi(y, z) d\gamma_y^-(z).$$

For every y take now $\gamma_y^+ \otimes \gamma_y^-$, which is a measure over $\Omega \times \Omega$. Define σ through

$$\int_{\Omega \times \Omega \times \Omega} \phi(x, y, z) d\sigma(x, y, z) := \int_{\Omega} d\rho(y) \int_{\Omega \times \Omega} \phi(x, y, z) d(\gamma_y^+ \otimes \gamma_y^-)(x, z).$$

It is easy to check that, for ϕ depending only on x and y , we have

$$\begin{aligned}
 \int_{\Omega^3} \phi(x, y) d\sigma &= \int_{\Omega} d\rho(y) \int_{\Omega \times \Omega} \phi(x, y) d(\gamma_y^+ \otimes \gamma_y^-)(x, z) \\
 &= \int_{\Omega} d\rho(y) \int_{\Omega} \phi(x, y) d\gamma_y^+(x) = \int \phi d\gamma^+.
 \end{aligned}$$

This proves $(\pi_{x,y})_{\#}\sigma = \gamma^+$ and the proof of $(\pi_{y,z})_{\#}\sigma = \gamma^-$ is analogous. $\square$ $\square$

Remark 5.6. All the analysis of this section can be performed on a general Polish space X instead of a subset $\Omega \subset \mathbb{R}^d$. The only differences are the definition of $\mathcal{P}_p(X)$ (we need to use $\mathcal{P}_p(X) := \{\mu \in \mathcal{P}(X) : \int d(x, x_0)^p d\mu(x) < +\infty\}$), and the fact that the first proof of the triangle inequality was specific to $\mathbb{R}^d$ (but the second works in general).

Definition 5.7. Given a Polish space X , for each $p \in [1, +\infty[$ we define its Wasserstein space of order p , $\mathbb{W}_p(X)$, as the space $\mathcal{P}_p(X)$, endowed with the distance W_p .

5.2 Topology induced by W_p

In this section we want to analyze the convergence in the space $\mathbb{W}_p$, and compare it to the notion of weak convergence. First of all, we recall what we mean by “weak convergence”: as we pointed out in Box 1.3 in Chapter 1, we use this term to denote the convergence in the duality with C_b , the space of bounded continuous functions (which is often referred to as narrow convergence), and we write $\mu_n \rightharpoonup \mu$ to say that μ_n converges in such a sense to μ .

We recall that another natural notion of convergence for measures is the one in duality with the space of functions vanishing at infinity

$$C_0(X) := \{\phi \in C(X) : \text{for all } \varepsilon > 0 \text{ there is } K \subset X \text{ compact s.t. } |\phi| < \varepsilon \text{ on } X \setminus K\}.$$

It is useful to point out some facts

- When X is compact, the spaces $C_0(X)$, $C(X)$ and $C_b(X)$ coincide.
- The condition defining $\phi \in C_0(X)$ means that ϕ tends to 0 “out of the compact subsets of X ”: this means at infinity, for $X = \mathbb{R}^d$, on $\partial\Omega$ when $X = \Omega \subset \mathbb{R}^d$ is open, and it means nothing if X is itself compact.
- For every $x_0 \in X$, the existence of functions $\phi \in C_0(X)$ with $\phi(x_0) \neq 0$ implies the existence of compact neighborhoods of x_0 . In particular, the space $C_0(X)$ is interesting only for locally compact spaces X and it only consists of the zero function whenever no point of X has compact neighborhoods (for instance, in infinite dimensional Banach spaces). Indeed, the duality result stating that $\mathcal{M}(X)$ is the dual of $C_0(X)$ is only true for locally compact spaces (see Box 1.3 in Section 1.1).

The following lemma (which is true in locally compact spaces, but we only state it for subsets of $\mathbb{R}^d$) gives a helpful criterion for weak convergence.

Lemma 5.8. *When all measures μ_n and μ are probabilities over a subset $\Omega \subset \mathbb{R}^d$, the convergence $\mu_n \rightharpoonup \mu$ coincides with the convergence in the duality with $C_0(\Omega)$.*

Proof. We only need to show that if we take $\psi \in C_b(\Omega)$, $\mu_n, \mu \in \mathcal{P}(\Omega)$ and we assume $\int \phi d\mu_n \rightarrow \int \phi d\mu$ for every $\phi \in C_0(\Omega)$, then we also have $\int \psi d\mu_n \rightarrow \int \psi d\mu$. If all the measures are probability, then we may add “for free” a constant C to ψ and, since ψ is bounded, we can choose C so that $\psi + C \geq 0$. Hence $\psi + C$ is the sup of an

increasing family of functions $\phi_n \in C_0$ (take $\phi_n = (\psi + C)\chi_n$, χ_n being an increasing family of cut-off functions⁴ with $\chi_n = 1$ on $B(0, n)$). Hence by semi-continuity we have $\int (\psi + C) d\mu \leq \liminf_n \int (\psi + C) d\mu_n$, which implies $\int \psi d\mu \leq \liminf_n \int \psi d\mu_n$. If the same argument is performed with $-\psi$ we obtain the desired convergence of the integrals. $\square$ $\square$

Once the weak convergence is understood, we can start from the following result.

Theorem 5.9. *If $\Omega \subset \mathbb{R}^d$ is compact, then $\mu_n \rightharpoonup \mu$ if and only if $W_1(\mu_n, \mu) \rightarrow 0$.*

Proof. Let us recall the duality formula, which gives for arbitrary $\mu, \nu \in \mathcal{P}(\Omega)$

$$W_1(\mu, \nu) = \max \left\{ \int_{\Omega} \varphi d(\mu - \nu) : \varphi \in \text{Lip}_1 \right\}.$$

Let us start from a sequence μ_n such that $W_1(\mu_n, \mu) \rightarrow 0$. Thanks to the duality formula, for every $\varphi \in \text{Lip}_1(\Omega)$ we have $\int \varphi d(\mu_n - \mu) \rightarrow 0$. By linearity, the same will be true for any Lipschitz function. By density, it will hold for any function in $C(\Omega)$. This shows that Wasserstein convergence implies weak convergence.

To prove the opposite implication, let us first fix a subsequence μ_{n_k} such that $\lim_k W_1(\mu_{n_k}, \mu) = \limsup_n W_1(\mu_n, \mu)$. For every k pick a function $\varphi_{n_k} \in \text{Lip}_1(\Omega)$ such that $\int \varphi_{n_k} d(\mu_{n_k} - \mu) = W_1(\mu_{n_k}, \mu)$. Up to adding a constant, which does not affect the integral, we can assume that the φ_{n_k} all vanish at a same point, and they are hence uniformly bounded and equicontinuous. By the Ascoli Arzelà Theorem we can extract a sub-subsequence uniformly converging to a certain $\varphi \in \text{Lip}_1(\Omega)$. By replacing the original subsequence with this new one we can avoid relabeling. We have now

$$W_1(\mu_{n_k}, \mu) = \int \varphi_{n_k} d(\mu_{n_k} - \mu) \rightarrow \int \varphi d(\mu - \mu) = 0,$$

where the convergence of the integral is justified by the weak convergence $\mu_{n_k} \rightharpoonup \mu$ together with the strong convergence (in $C(\Omega)$) $\varphi_{n_k} \rightarrow \varphi$. This shows that $\limsup_n W_1(\mu_n, \mu) \leq 0$ and concludes the proof. $\square$ $\square$

Theorem 5.10. *If $\Omega \subset \mathbb{R}^d$ is compact and $p \in [1, +\infty[$, in the space $\mathbb{W}_p(\Omega)$ we have $\mu_n \rightharpoonup \mu$ if and only if $W_p(\mu_n, \mu) \rightarrow 0$.*

Proof. We have already proved this equivalence for $p = 1$. For the other values of p , just use the inequalities

$$W_1(\mu, \nu) \leq W_p(\mu, \nu) \leq C W_1(\mu, \nu)^{\frac{1}{p}},$$

that give the equivalence between the convergence for W_p and for W_1 . $\square$ $\square$

We can now pass to the case of unbounded domains. From the fact that the distance W_p does not really depend on the ambient domain, we can simply consider the case $\Omega = \mathbb{R}^d$.

⁴The existence of cut-off functions, i.e. a sequence of continuous compactly supported functions converging pointwisely to 1, is indeed peculiar to the case of locally compact spaces.

Theorem 5.11. *In the space $\mathbb{W}_p(\mathbb{R}^d)$, we have $W_p(\mu_n, \mu) \rightarrow 0$ if and only if $\mu_n \rightharpoonup \mu$ and $\int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$.*

Proof. Consider first a sequence μ_n which is converging to μ in $\mathbb{W}_p(\mathbb{R}^d)$. It is still true in this case that

$$\sup \left\{ \int \phi d(\mu_n - \mu) : \phi \in \text{Lip}_1 \right\} \rightarrow 0,$$

which gives the convergence when testing against arbitrary Lipschitz functions. Note that Lipschitz functions are dense (for the uniform convergence) in the space $C_0(\Omega)$ (while it is not necessarily the case for $C_b(\Omega)$). This is enough to prove $\mu_n \rightarrow \mu$, thanks to Lemma 5.8.

To obtain the other condition, namely $\int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$ (which is not a consequence of the weak convergence, since $|x|^p$ is not bounded), it is sufficient to note that

$$\int |x|^p d\mu_n = W_p^p(\mu_n, \delta_0) \rightarrow W_p^p(\mu, \delta_0) = \int |x|^p d\mu.$$

We need now to prove the opposite implication. Consider a sequence $\mu_n \rightharpoonup \mu$ satisfying also $\int |x|^p d\mu_n \rightarrow \int |x|^p d\mu$. Fix $R > 0$ and consider the function $\phi(x) := (|x| \wedge R)^p$, which is continuous and bounded. We have

$$\begin{aligned} \int (|x|^p - (|x| \wedge R)^p) d\mu_n &= \int |x|^p d\mu_n - \int \phi d\mu_n \\ &\rightarrow \int |x|^p d\mu - \int \phi d\mu = \int (|x|^p - (|x| \wedge R)^p) d\mu. \end{aligned}$$

Since $\int (|x|^p - (|x| \wedge R)^p) d\mu \leq \int_{B(0,R)^c} |x|^p d\mu$ and $\int_{\mathbb{R}^d} |x|^p d\mu < \infty$, it is possible to choose R so that

$$\int (|x|^p - (|x| \wedge R)^p) d\mu < \varepsilon/2$$

and hence one can also guarantee that $\int (|x|^p - (|x| \wedge R)^p) d\mu_n < \varepsilon$ for all n large enough.

We use now the inequality $(|x| - R)^p \leq |x|^p - R^p = |x|^p - (|x| \wedge R)^p$ which is valid for $|x| \geq R$ (see Lemma 5.12 below) to get

$$\int (|x| - R)^p d\mu_n < \varepsilon \text{ for } n \text{ large enough, and } \int (|x| - R)^p d\mu < \varepsilon.$$

Consider now $\pi_R : \mathbb{R}^d \rightarrow \overline{B(0,R)}$ defined as the projection over $\overline{B(0,R)}$. This map is well-defined and continuous and is the identity on $B(0,R)$. Moreover, for every $x \notin B(0,R)$ we have $|x - \pi_R(x)| = |x| - R$. We can deduce

$$W_p(\mu, (\pi_R)_\# \mu) \leq \left(\int (|x| - R)^p d\mu \right)^{\frac{1}{p}} \leq \varepsilon^{\frac{1}{p}},$$

$$W_p(\mu_n, (\pi_R)_\# \mu_n) \leq \left(\int (|x| - R)^p d\mu_n \right)^{\frac{1}{p}} \leq \varepsilon^{\frac{1}{p}}.$$

Note also that, due to the usual composition of the weak convergence with continuous functions, from $\mu_n \rightharpoonup \mu$ we also infer $(\pi_R)_\# \mu_n \rightharpoonup (\pi_R)_\# \mu$. Yet, these measures are all concentrated on the compact set $\overline{B(0, R)}$ and here we can use the equivalence between weak convergence and W_p convergence. Hence, we get

$$\begin{aligned} \limsup_n W_p(\mu_n, \mu) &\leq \limsup_n (W_p(\mu_n, (\pi_R)_\# \mu_n) + W_p((\pi_R)_\# \mu_n, (\pi_R)_\# \mu) + W_p(\mu, (\pi_R)_\# \mu)) \\ &\leq 2\varepsilon^{\frac{1}{p}} + \lim_n W_p((\pi_R)_\# \mu_n, (\pi_R)_\# \mu) = 2\varepsilon^{\frac{1}{p}}. \end{aligned}$$

As the parameter $\varepsilon > 0$ is arbitrary, we get $\limsup_n W_p(\mu_n, \mu) = 0$ and the proof is concluded. $\square$

Lemma 5.12. *For $a, b \in \mathbb{R}_+$ and $p \geq 1$ we have $a^p + b^p \leq (a + b)^p$.*

Proof. Suppose without loss of generality that $a \geq b$. Then we can write $(a + b)^p = a^p + p\zeta^{p-1}b$, for a point $\zeta \in [a, a + b]$. Use now $p \geq 1$ and $\zeta \geq a \geq b$ to get $(a + b)^p \geq a^p + b^p$. $\square$

Remark 5.13. How to handle more general metric spaces X instead of subsets of $\mathbb{R}^d$? First, we notice that Lemma 5.8 does not hold in general (for instance, it is trivially false in infinite dimensional Banach spaces). Also, the problem of the density of Lipschitz functions in the space of continuous functions (even if X is compact) appears. Indeed, in $\mathbb{R}^d$ the fact that smooth functions are dense is easy to obtain by convolution, a tool which is not available in general spaces. To overcome this difficulty, one can prove the following statement, which is enough for our scopes: “if we have convergence testing against arbitrary Lipschitz functions, then we also have convergence testing against arbitrary functions in C_b ” (this is left as an exercise, see Ex(31)). Hence, in what concerns Theorems 5.9 and 5.10, everything can be translated into the framework of Polish spaces (with no need of local compactness). On the other hand the proof that we gave of Theorem 5.11 was somehow specific to $\mathbb{R}^d$: the key point was the existence of a continuous retraction π_R from the whole space to a compact subset (the ball $\overline{B(0, R)}$), with a bound on the distance between x and $\pi_R(x)$. The proof could be adapted to more general spaces, with ad-hoc assumptions, but we prefer to refer the reader to [17, 10] for the general case.

5.3 Lipschitz curves in $\mathbb{W}_p$ and the continuity equation

In this section we analyze some properties of Lipschitz and absolutely continuous curves in the space $\mathbb{W}_p$. In order to do that, we need some simple elements from analysis in metric spaces. The reader who wants to know more can have a look, for instance, at [13].